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King Saud University
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Course: IT461 Practical Machine Learning

Project Proposal:
Detecting Hallucinations in Large Language Model Responses

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Table of contents

1. Motivation 3

.2 Background 4

2.1 Large Language Models (LLMs) and the Hallucination Phenomenon 4

2.2 Context Grounding 4

2.3 Supervised Binary Classification for Text Verification 5

2.4 Text Representation 5

3. Related Works 5

4. Dataset 6

5. Contributions 8

6. References 9

Table of Figures

Figure1 : Proposed flowchart of the hallucination detection system. 4

Figure2: Representative sample from the HaluEval QA subset 8

1. Motivation

Problem Significance and Motivation:

Large Language Models (LLMs) are increasingly used to generate answers for many different tasks. Although these models can produce fluent and convincing responses, they may sometimes generate information that is incorrect, unsupported, or different from the provided source. This problem is known as hallucination and can reduce the reliability of LLM based systems.

Detecting hallucinations is important because incorrect information can mislead users, especially when LLMs are used in applications where factual accuracy is important. Therefore, there is a need for an automated method that can check whether a generated answer is supported by reliable source information.

In this project, we investigate hallucination detection as a supervised machine learning classification problem. The model uses the question, the provided knowledge, and a candidate answer to determine whether the answer is faithful to the source or contains hallucinated information. We will use the HaluEval QA dataset to train and evaluate supervised machine learning models for distinguishing between faithful and hallucinated responses. The models will be compared on a held-out test set using accuracy, precision, recall, and F1-score, with particular attention to their ability to detect hallucinated responses.

Target Task: Supervised Binary Classification.

Inputs & Outputs:

The system receives the question, the reference knowledge, and one candidate answer. It then predicts whether the candidate's answer is faithful to the provided knowledge or hallucinated.

Inputs:

1. **Knowledge (Context):** The factual reference information used to verify the answer.
2. **Question:** The question that the LLM is expected to answer.
3. **Candidate Answer:** One answer that needs to be classified. It can be either the original faithful answer or the hallucinated answer from the HaluEval dataset.

Outputs:

The model produces a binary classification label:

- **(0) Faithful:** The candidate answer is consistent with and supported by the provided knowledge.
- **(1) Hallucinated:** The candidate answer contains information that is unsupported or inconsistent with the provided knowledge.

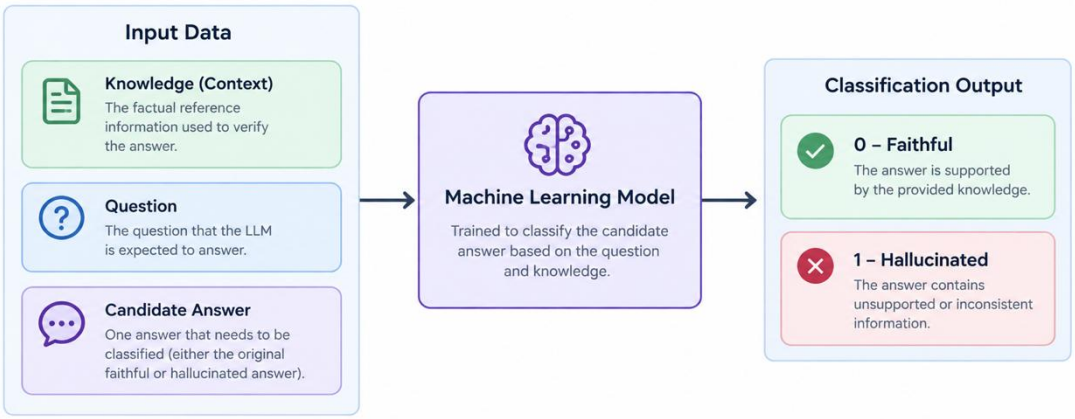


Figure 1 : Proposed flowchart of the hallucination detection system.

2. Background

Key Concepts

To fully understand the scope and methodology of this project, several foundational concepts in Machine Learning and Natural Language Processing (NLP) must be established:

2.1 Large Language Models (LLMs) and the Hallucination Phenomenon

Large Language Models generate text based on probabilistic patterns learned from massive corpora. However, a major limitation in these models is "hallucination" a phenomenon where the model produces responses that sound fluent and plausible but are factually incorrect, unsupported, or contradictory to the provided source text. Detecting these discrepancies is critical for building trustworthy AI systems.

2.2 Context Grounding

Context grounding refers to evaluating a generated response based on provided reference information. In this project, the source context is used as the factual reference to determine whether the generated answer is supported by the source or contains hallucinated information.

2.3 Supervised Binary Classification for Text Verification

This project frames hallucination detection as a supervised binary classification task. Using a labeled dataset, machine learning algorithms are trained to map input features (derived from the query, context, and response) to one of two distinct target classes: *Faithful* (factually consistent with the source) or *Hallucinated* (containing ungrounded or contradictory information).

2.4 Text Representation

Machine learning models require textual data to be converted into numerical representations before classification. In this project, the question, source context, and generated answer are processed and represented numerically so that a supervised classifier can determine whether the response is faithful or hallucinated.

3. Related Works

1. HaluEval Benchmark (Li et al., 2023):

- *What they did:* Li et al. [1] introduced HaluEval, a large-scale hallucination evaluation benchmark containing 35,000 human-annotated and automatically generated samples across question answering, dialogue, and summarization tasks.
- *What they found:* They demonstrated that even powerful LLMs (like ChatGPT) struggle to reliably recognize hallucinations in generated text without external knowledge or reasoning steps [1].
- *How our project differs:* While HaluEval provides a benchmark for evaluating hallucinations, our project uses its QA dataset to train a supervised binary classification model that determines whether a response is faithful or hallucinated.

2. SelfCheckGPT (Manakul et al., 2023):

- *What they did:* Manakul et al.[2] proposed SelfCheckGPT, a zero-resource, black-box approach to detect hallucinations by checking the consistency across multiple stochastically sampled responses from an LLM.
- *What they found:* Their method showed high correlation in assessing sentence-level and passage-level factuality without relying on external databases[2].
- *How our project differs:* Unlike SelfCheckGPT, which requires multiple generated responses for consistency checking, our project uses a provided

reference context and supervised binary classification to determine whether a candidate response is faithful or hallucinated.

3. Taxonomy of Hallucinations in NLG (Ji et al., 2023):

- *What they did:* Ji et al.[3] provided a comprehensive survey classifying the types, causes, and mitigation strategies of hallucinations in Natural Language Generation (NLG).
- *What they found:* They categorized hallucinations into intrinsic (conflicting with source) and extrinsic (unsupported by source) types, emphasizing the difficulty of factual alignment[3].
- *How our project differs:* While their work provides a theoretical taxonomy of hallucinations, our project focuses on automatically classifying generated responses as faithful or hallucinated based on the provided source context.

4. Probing LLM Truthfulness (Azaria & Mitchell, 2023):

- *What they did:* Azaria and Mitchell [4] investigated whether the internal states of language models can determine if a statement is true or false by training a classifier on the model's hidden states.
- *What they found:* Simple classifiers trained on model activations could effectively distinguish between true and false assertions[4].
- *How our project differs:* Rather than examining the internal hidden states of language models, our project uses the provided reference context and labeled data to train a supervised binary classifier for hallucination detection.

4. Dataset

Description & Source: We will use the publicly available question-answering (QA) subset of the HaluEval dataset [1], [5]. HaluEval is a benchmark developed to evaluate hallucinations in Large Language Models (LLMs) and contains 35,000 examples across multiple tasks. The selected QA subset contains 10,000 records derived from HotpotQA. Each record includes four textual attributes: knowledge, question, right_answer, and hallucinated_answer. For binary classification, each record will be converted into two instances: the right answer will be labeled 0 (faithful), while the hallucinated answer will be labeled 1 (hallucinated). Therefore, the final dataset will contain 20,000 balanced instances. The model input consists of

the knowledge, question, and one candidate answer, while the target variable is the hallucination label.

Justification: HaluEval was selected because it directly supports the project's objective of detecting hallucinations in LLM responses. It provides both a faithful and a hallucinated answer for the same question and reference knowledge, enabling a fair comparison between the two classes. The dataset is sufficiently large, balanced, publicly available, and suitable for reproducible supervised learning experiments.

Summary Statistics:

Characteristic	Value
Dataset	HaluEval QA Subset
Original Records	10,000
Number of Attributes	4
Attribute Type	Textual
Model Instances after Restructuring	20,000
Number of Classes	2
Class Balance	10,000 faithful and 10,000 hallucinated responses (50% each)
Missing Values	0

Table 1. Dataset statistics of the HaluEval QA subset.

Data Sample: Figure 2 presents an actual record from the selected QA subset. It shows the reference knowledge and question together with a faithful response labeled 0 and a hallucinated response labeled 1. The example illustrates how each original record will be converted into two labeled instances for supervised learning.



Figure 2 Representative sample from the HaluEval QA subset

5. Contributions

- **Aisha Fahad Alonizy:** Wrote the Motivation section, designed the architectural schematic for the end-to-end classification system (Figure 1), created the cover page, and formatted the document according to the requirements.
- **Remas Almutairi:** Researched the key concepts and wrote the Background section, including the summaries of the related works. Standardized the section headings and prepared the Table of Contents and Table of Figures.
- **Dalal AlGhumlas:** Wrote the Dataset section, prepared the summary statistics (Table 1) and data sample (Figure 2), and compiled the IEEE references list.

6. References

[1] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, and J.-R. Wen, "HaluEval: A large-scale hallucination evaluation benchmark for large language models," *arXiv preprint arXiv:2305.11747*, 2023. [Online]. Available: HaluEval on arXiv

[2] P. Manakul, A. Liusie, and M. J. F. Gales, "SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models," *arXiv preprint arXiv:2303.08896*, 2023. [Online]. Available: SelfCheckGPT on Hugging Face Papers

[3] Z. Ji *et al.*, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023. [Online]. Available: Semantic Scholar

[4] A. Azaria and T. Mitchell, "The internal state of an LLM knows when it's lying," *arXiv preprint arXiv:2304.13734*, 2023. [Online]. Available: Azaria & Mitchell on arXiv

[5] RUCAIBox, "HaluEval: A hallucination evaluation benchmark for large language models," GitHub. [Online]. Available: https://github.com/RUCAIBox/HaluEval

[Accessed: Sep. 6, 2026].